

Can a carbon-aware VCG rule allocate shared GPU-hours better than first come, first served?

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KEY MESSAGE

A published, carbon-aware VCG rule beats or ties all 720 FCFS arrival orders on value net of carbon (19.2 vs. 16.96 mean), cuts emissions per unit of value (0.72 vs. 0.81 kg), and keeps truthful reporting optimal at every carbon weight tested, but serves fewer teams (4 vs. 4.93).

01 Question & model

Setting & verified gap

Setting: AI demand outpaces compute; queues ignore both value and carbon.
Gap: FAIR-Compute [5] shows that value-blind scheduling loses about a third of scientific value, but it does not quantify carbon and leaves truthful-reporting mechanisms to future work. We build that piece at an exactly solvable scale.

Strategic model

Players: six teams A–F; each privately knows $v_i \in \{3, 6, 9\}$ and sends one sealed report r_i (static game, incomplete information).
Public: request d_i , emissions $e_i = 0.24 d_i$ kg CO₂e, capacity $K = 100$, $\lambda = 0.5$.
Rule: pick the feasible group S^* maximizing $\sum(r_i - \lambda e_i)$ over all 64 groups; each winner pays the score the other teams lose because it is selected.
Payoff: $u_i = v_i - \lambda e_i - p_i$ if selected, else 0.
Solution: truthful $r_i = v_i$ is dominant (DSIC) [1–4]: the price never depends on a team's own report.
Analogy stops: credits are not money; values are never verified.

Reduced payoff matrix: Team C vs. Team E

(u_C, u_E)	E truthful ($r = 6$)	E underreports ($r = 4$)
C truthful ($r = 3$)	(0, 1.2)	(0.8, 0)
C overreports ($r = 4.5$)	(−1.2, 0)	(0.8, 0)

Simplification: only C and E choose; A, B, D, F report truthfully. Truth is weakly dominant for both, and C's overreport pushes the truthful E out.

Three lenses

Game theory: truthful reporting is each team's best strategy, whatever others report.
Social choice: value net of carbon under capacity; the cost is fewer teams served (4 vs. 4.93).
Mechanism design: a published score rule with VCG prices replaces the queue; λ moves prices, not the winners or the best strategy.

02 Design & evidence

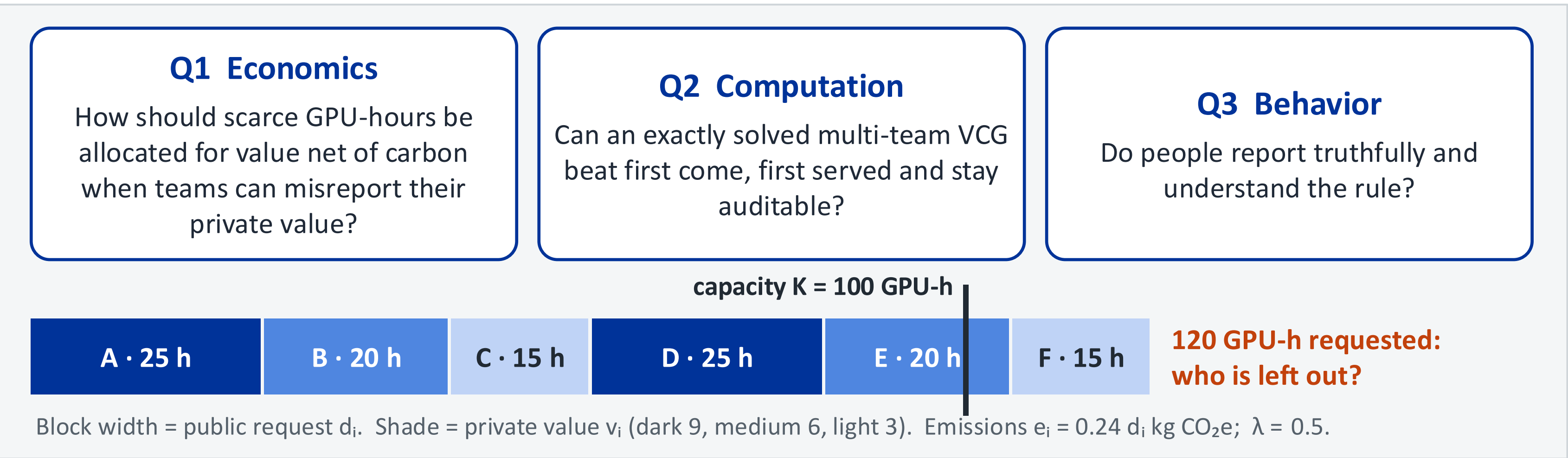


Figure 1. Research questions and setting. Six teams request 120 GPU-hours from a pool of 100. Values are private; requests and emissions are public. Q1 and Q2 are answered by the model and its exact computation, Q3 by play in the Hugging Face Space.

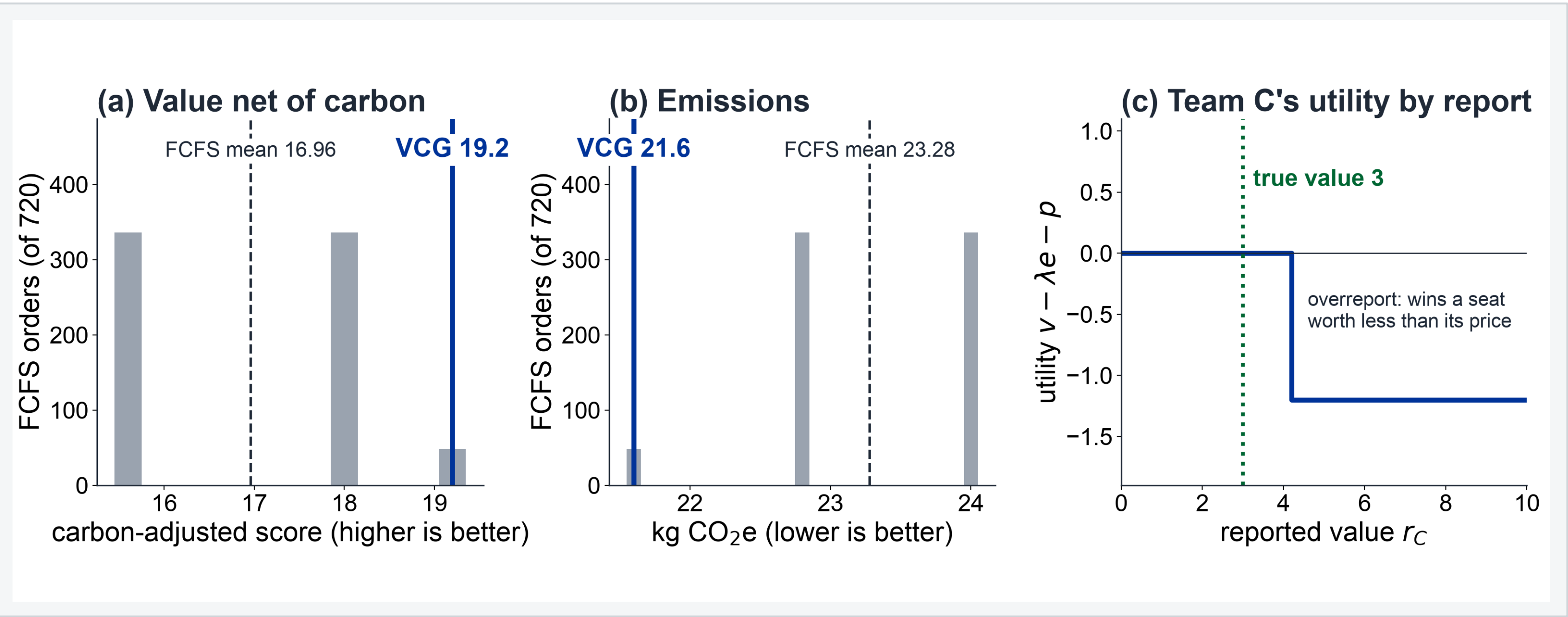


Figure 2. FCFS outcomes across all 720 arrival orders vs. VCG (a, b), and Team C's utility for every report (c).
Result: VCG picks A, B, D, E (value 30, 21.6 kg CO₂e); no FCFS order beats it, but FCFS serves more teams on average.
Parameter change: for λ from 0.1 to 1 the same group wins, the price per winner falls from 5.28 to 0, and truth stays optimal. With two GPU types (H100, A100) the same teams win and emissions fall from 21.6 to 16.6 kg.
Verification: python colab/run_simulation.py reproduces the outputs; 7 unit tests pass; notebook 9/9 checks PASS; commit 8a656a4.

03 Impact & dialogue

Practical impact: a real-world case

Case: a university's shared GPU cluster replaces queue order with a published, re-runnable rule and a carbon weight λ set by the institution; no adoption yet.
Benefits / risks: compute goes to value net of carbon and every allocation is auditable; fewer teams are served, and a heavy λ penalizes compute-heavy research.
Who needs what: academia, a transparent pilot; industry, robustness at hundreds of teams; public institutions, a re-runnable audit.
Correction: rule, inputs, and code are public, so any team can re-run and challenge an allocation.

Behavioral evidence: five self-plays (exploratory)

Measure	Benchmark	Observed
Truthful reports ($r = v$)	5 / 5	2 / 5
Overreports	0	3
Selection predicted correctly	—	3 / 5
Plays where a misreport raised utility	0	0

Agreement: no misreport paid off; one overreport bought a seat for 36 credits and ended at $u = -3.6$.
Gap: three of five players overreported. Stated reasons: wanting to be selected, and misunderstanding the rule.
Integration: the Space's what-if curve exposed that utility $v_i - p_i$ rewards overreporting; each team now bears its own carbon.

Peer review, limits & next test

Why $\lambda = 0.5$? Sweep 0 to 1: prices change, the best strategy does not.
One GPU type? Two-pool extension: same winners, emissions −23%.
No play data? Five recorded self-plays by the authors.
One instance? Truth also holds on 100 random instances; the FCFS comparison over random draws is next.
Limits: λ and 0.24 kg/GPU-h are assumptions; true values cannot be verified; authors played against simulated rivals.

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Selected references

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[2] Clarke, E. H. (1971). Multipart pricing of public goods. *Public Choice* 11, 17–33. doi:10.1007/BF01726210
[3] Groves, T. (1973). Incentives in teams. *Econometrica* 41(4), 617–631. doi:10.2307/1914085
[4] Harsanyi, J. C. (1968). Games with incomplete information played by Bayesian players, II. *Mgmt. Sci.* 14(5), 320–334. doi:10.1287/mnsc.14.5.320
[5] Zachariadis, K. E., et al. (2026). FAIR-Compute: A roadmap for fair and efficient allocation of federated digital research infrastructure. arXiv:2607.28290

OPEN MATERIALS

QR codes: left Hugging Face Space, right GitHub repository.

Hugging Face: https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Team_A
GitHub: <https://github.com/dku-comsci-econ206-Autumn2026/FP05-PS2-Team-A> (commit 8a656a4)
Colab: <https://colab.research.google.com/drive/14eo8a2HWUbbBtv-m-ja4xRiL4V6lwttun>

